VIBRATION EXPERIMENT

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Course Code:

- SPC 408

Work Distribution

Names	Free Vibration	Forced Vibration
Asmaa Sabry	• Describe the experimental setup and its main components. • Describe the steps of the experimental procedure. • Describe the mathematical model of the system and its time response. • Data Reading in the lab	• Describe the experimental setup and its main components. • Describe the steps of the experimental procedure. • Describe the mathematical model of the system and its time response. • Data Reading in the lab
Raed Asham	• Determine the beam's equivalent mass and stiffness. • Use the measured time response to determine the experimental system's logarithmic decrement, damping ratio, damped natural frequency, and natural frequency. • Determine the theoretical natural frequency, damped natural frequency, and equivalent damping coefficient of the system. • Plot the measured time response and the corresponding theoretical response.	• Plot the measured displacement amplitude ratio versus the frequency ratio ω/ωn. • Plot the theoretical displacement amplitude ratio versus the frequency ratio ω/ωn.
Iman Mosaad	• Compare the experimental and theoretical results • Comment on the results and identify any sources of error • Data Reading in the lab	• Compare between the experimental and theoretical results. • Comment on the results and identify any sources of error. • Data Reading in the lab

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Abstract

This experiment is induced by the response of the free and forced vibration of the cantilever beam fixed to the shaker after it was hit by initial disturbance. This experiment aims to identify the natural frequency, the damping ratio, and the decrement of the beam, and also to study the effect of changing the excitation on its behaviour when resonating. Then learn about the relationship between the displacement amplitude and the frequency ratio. Collect the data through the accelerometer and then analyse it through the data collection and analysis system and obtain the damping ratio and... amplitude of the beam, and then the data is compared and the experimental and theoretical results are verified. Through the outline, we can know the results and procedures for each of the two experiments and clarify the theoretical and experimental results

Introduction

When we observe anything around us, we find that vibration is present in everything that is related to geometry. Where vibration occurs in vehicles, buildings, machines, etc. Therefore, we must learn how to deal with it, otherwise the situation becomes very dangerous, leading to damage and destruction, especially when resonance occurs. These experiments study the effect of vibration on the cantilever beam, which is the free vibration and forced vibration, so we find in the first experiment Concerning free vibration, the beam was moved and left to vibrate freely, and information was obtained about the natural frequency and damping characteristics. In the second experiment on forced vibration, we find that the beam vibrates at different frequencies and observe the movement resulting from that and how it changes, also when resonating. In these experiments, the experimental results are linked to the theoretical results and compared between them in order to identify the behaviour of the vibration. This report explains the setup of the experiment **(fig. 1, 2, 3)**, the procedures and the results for both cases and compares the experimental results with the theoretical results.

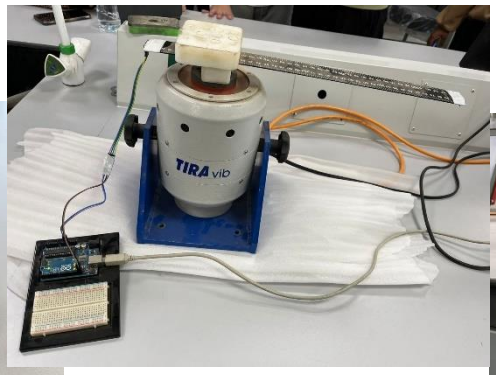


Figure 1: The Beam attached on the Shaker



Figure 2: Function Generator



Figure 3: DC power supply

Free Vibration Experiment

1. Experimental Setup

- **Accelerometer:** It is a sensor located in the beam to detect the vibrational moves
- **Cantilever Beam:** The beam which is fixed at one end. It vibrates when disturbed and shows the SDOF system.
- **Data Recorder:** The device that help to manage the collected data that comes from the accelerometer.
- **Computer/Software:** The devise that help to visualize the data collected with the info.

2. Experimental Procedure

First, the setup is prepared by making the shaker and accelerometer connected to the cantilever beam that has been installed well. Then you make sure that the accelerometer is connected to the device that collects the data, and then they are connected to the computer, after which the dimensions of the beam are recorded and the material properties are taken. . After that, the information collection device is turned on and it is ensured that the software is ready to start performing the experiment and taking data. After that, a small distance is applied to the beam from its free end, and we leave it to vibrate and move freely. From here, the data is recorded by the computer, and this recorded data is analysed and parameters such as natural frequency and damping are obtained.

3. Mathematical Model & Time Response

➢ Assuming the setup model to be a SDOF damped vibration system.

The equation that describes the motion is:

$$\boldsymbol{m\ddot{x} + c\dot{x} + kx = 0}$$

Where:

$\boldsymbol{m}$ stands for the mass

$\boldsymbol{c}$ stands for the damping coefficient

$\boldsymbol{k}$ stands for the stiffness

$\boldsymbol{x(t)}$ stands for the displacement as a function of time

Then the solution of the previous equation is:

$$\boldsymbol{x(t) = X_0 e^{-\zeta\omega_n t}\cos(\omega_d t + \phi)}$$

Where:

X_0 stands for the initial displacement

ζ stands for the damping ratio

ω_n stands for the natural frequency

ω_d stands for the damped natural frequency

ϕ stands for the phase angle

4. Equations and Calculations

- Equivalent mass of the beam: $m_{eq} = \frac{\rho bhL}{2}$

  Where:
  - ρ is the beam material density. For steel: $\rho = 7750\ \text{kg/m}^3$
  - b is the beam's width. $b = 38.6\ \text{mm}$
  - h is the beam's thickness. $h = 1.4\ \text{mm}$
  - L is the beam's length. $L = 28.5\ \text{cm}$
    - Hence, $m_{eq} = 0.0597\ \text{kg}$
- Equivalent stiffness of the beam: $k_{eq} = \frac{3EI}{L^3}$

  Where:
  - E is the Modulus of elasticity of the material. For steel: $E = 200\text{GPa}$
  - I is the Moment of inertia of the beam's cross-section. $I = \frac{1}{12}bh^3$

    $(b = 38.6\text{mm}, h = 1.4\ \text{mm}) \rightarrow \quad I = 8.827 \times 10^{-12}\ \text{m}^4$
  - L is the beam's length. $L = 28.5\ \text{cm}$
    - Hence, $k_{eq} = 228.79\ \text{N/m}$

Measured Values:

- **Logarithmic Decrement:** $\delta = \frac{1}{n}\ln\left(\frac{x_0}{x_n}\right)$

  Where:
  - x_0 the value at an initial peak.
  - x_n the value of the peak after n cycles.

  By taking a section from the graphs of the three trials:

	x_o	x_n	n	δ
First Trial	4.962285	2.161709	12	0.06925
Second Trial	4.047446	1.91084		0.06255
Third Trial	4.468872	2.107017		0.06266

  The δ value from the first trial may seem outlier; hence, neglected. By taking the average between the second and third trials, we conclude $\delta = 0.06261$.

- **Damping Ratio:** $\zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}}$

  Substituting with the previously calculated value of δ:

  $\zeta = 9.964 \times 10^{-3}$

❖ **Damped Natural Frequency:** $\omega_d = \frac{2\pi}{T_d}$

Where:

- T_d the time period of the damped oscillation. $T_d = 0.121773$ sec
  - Hence, $\omega_d = 51.6$ rad/sec

❖ **Natural Frequency:** $\omega_n = \frac{\omega_d}{\sqrt{1-\zeta^2}}$

Substituting with the previously calculated values of ζ and ω_d: $\omega_n = 51.6$ rad/sec

➢ Theoretical Values:

❖ **Natural Frequency:** $\omega_n = \sqrt{\frac{k_{eq}}{m_{eq}}}$

Substituting with the previously calculated values of k_{eq} and m_{eq}: $\omega_n = 61.906$ rad/sec

❖ **Damped Natural Frequency:** $\omega_d = \omega_n\sqrt{1-\zeta^2}$

Substituting with the previously calculated values of ζ (9.964×10^{-3}) and ω_n: $\omega_d = 61.903$ rad/sec

❖ **Damping Coefficient:** $c = 2\zeta\sqrt{m_{eq}k_{eq}}$

Where:

- $m_{eq} = 0.0597$ kg
- $k_{eq} = 225.24$ N/m
- $\zeta = 9.964 \times 10^{-3}$
  - Hence, $c = 0.0736$ kg/sec

5. Plots

Since the damping ratio is less than one ($\zeta \ll 1$), the system is considered underdamped. Hence, the equation used for plotting is:

$$x = e^{-(\zeta\omega_n)t}\left(A\sin\left(\sqrt{1-\zeta^2}\omega_n t\right) + B\cos\left(\sqrt{1-\zeta^2}\omega_n t\right)\right),$$

where $A = \frac{\dot{x}(0)+\zeta\omega_n x(0)}{\omega_n\sqrt{1-\zeta^2}}$ and $B = x(0)$

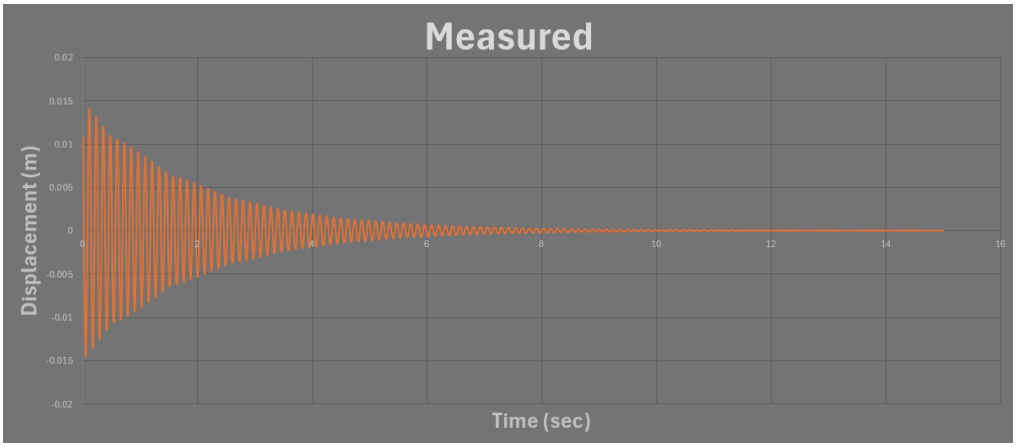
Both measured and theoretical approaches have:

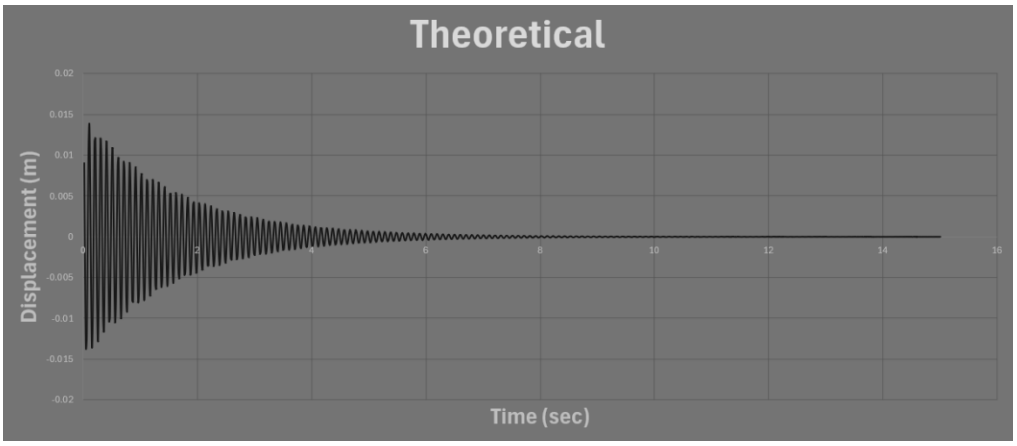
- $x(0) = 1.5$ cm **(given)**
- $\dot{x}(0) = 0$ **(given)**
- $\zeta = 9.964 \times 10^{-3}$

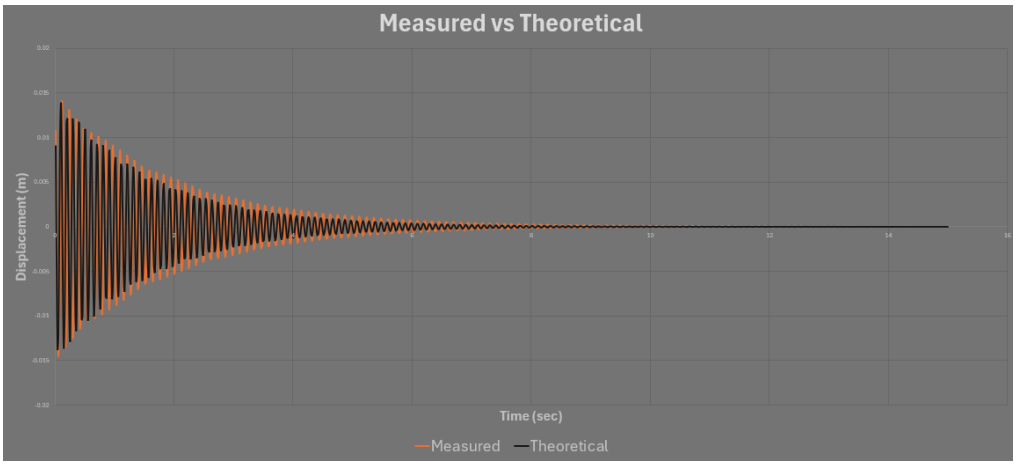
The difference between the measured and theoretical is in the natural frequency.

- For Measured: $\omega_n = 51.6$ rad/sec
- For Theoretical: $\omega_n = 61.906$ rad/sec

➢ Substituting in the given equation with the previous values and for $0 \le t \le 15$ sec with $dt = 0.015$, the plots are:







(For more details and data, see attached file "*Free Vibration Plots*")

6. Experimental and Theoretical Results Comparison

- The experimentally natural frequency is equal to $\boldsymbol{\omega_n = 51.6\ rad/sec}$ and the theoretical value is equal to $\boldsymbol{\omega_n = 61.906\ rad/sec}$ which means that the experimental value is **16.65%** lower than the theoretical value.
- Same with the values of experimented and theoretical values of damped natural frequency $\boldsymbol{\omega_d}$**.**
- Damping ratio is the same for both experimental and theoretical**.**
- The amplitude of oscillations in the experimental decays faster than the theoretical one.

7. Comments on Results & Error Sources

- The natural frequency from the experimental results is slightly lower than the theoretical value because of the damping presence and energy dissipation as for the theoretical calculations idealized assumptions is considered such as uniform material properties and neglection of minor nonlinearities.
- The theoretical results provide more accurate approximation. Thus, it deviates from the experimental results because of the idealized assumptions.
- These differences focus on the effect of damping, friction, and material properties in real systems.

These discrepancies result from some source of errors in the real system which could be:

- **Measurement Errors:** errors of measuring the time intervals, displacement, and frequency using the experimental system.
- **Material and Structural Imperfections:** nonideal material properties like stiffness and damping. Any small difference in the setup impact in the setup vibration.
- **Energy Losses:** energy losses due to air resistance, friction, or internal material damping.
- Initial values were approximated.
- The system was idealized to be a SDOF system.

Forced Vibration Experiment

1. Experimental Setup

- **Shaker:** This makes the harmonic excitation to create vibration in the Beam.
- **Accelerometer:** It is a sensor located in the beam to detect the amplitude
- **Cantilever Beam:** The beam which is fixed at one end. It vibrates when disturbed and shows the SDOF system.
- **Data Recorder:** The device that help to manage the collected data that comes from the accelerometer then transmits it to the computer.
- **Computer/Software:** The devise that help to visualize the data collected with the info.
- **Function Generator:** This adjusts the amplitude and the frequency and generate the harmonic signal for the shaker

2. Experimental Procedure

First, the setup is prepared by making the shaker and accelerometer connected to the cantilever beam at its free end. Then you make sure to connect the accelerometer to the device that collects the data, and then they are connected to the computer. After that, make sure that the frequency function generator is connected to the next control. From the shaker, taking into account starting from a low value. After that, the excitation frequency is gradually increased and the beam is observed, after which the frequency is recorded at the maximum value of the amplitude at the resonance. After that, the information collection device is turned on and it is ensured that the software is ready to start performing the experiment and taking data. After that, the amplitude data for each excitation frequency is recorded. After that, the data is recorded so that the amplitude and frequency ratio are plotted.

3. Mathematical Model & Time Response

- Assuming the setup model to be a SDOF damped forced vibration system.

The equation that describes the motion is:

$$\boldsymbol{m\ddot{x} + c\dot{x} + kx = F_0 \cos(\omega t)}$$

Where:

$\boldsymbol{F_0}$ stands for the amplitude of the harmonic force

$\boldsymbol{\omega}$ stands for the excitation frequency

$\boldsymbol{m}$ stands for the mass

$\boldsymbol{c}$ stands for the damping coefficient

$\boldsymbol{k}$ stands for the stiffness

$\boldsymbol{x(t)}$ stands for the displacement as a function of time

Then the solution of the previous equation is:

$$X(\omega) = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}$$

And the amplitude is calculated by:

$$\frac{X(\omega)}{Y} = \frac{1}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}}$$

Where:

Y stands for the initial displacement

ζ stands for the damping ratio

r stands for the frequency ratio $\left(r = \frac{\omega}{\omega_n}\right)$

The time response can get from this equation:

$$x(t) = X(\omega)\cos(\omega t - \phi)$$

Where:

ϕ stands for the phase angle

4. Equations and Calculations

From (*Equations and Calculations part in the free vibration system*):

- Equivalent mass of the beam: $m_{eq} = 0.0597\ \text{kg}$
- Equivalent stiffness of the beam: $k_{eq} = 228.79\ \text{N/m}$
- Beam theoretical natural frequency: $(\omega_n)_{theo} = 61.906\ \text{rad/sec}$
- Beam measured natural frequency: $(\omega_n)_{meas} = 51.6\ \text{rad/sec}$
- Damping ratio: $\zeta = 9.964 \times 10^{-3}$

➢ Measured Values:

- ω is measured for each frequency (given in **Hz**) using $\omega = 2\pi f$
- $\zeta = 9.964 \times 10^{-3}$
- $(\omega_n)_{meas} = 51.6\ \text{rad/sec}$
- **X** and **Y** were extracted by the Accelerometer for each frequency.
- r is calculated using the equation $r = \frac{\omega}{\omega_n}$ for each ω.
  - **(For the results and data of the two trials, see appendices "*Forced Vibration Measured Data Trial 1*" and "*Forced Vibration Measured Data Trial 2*")**

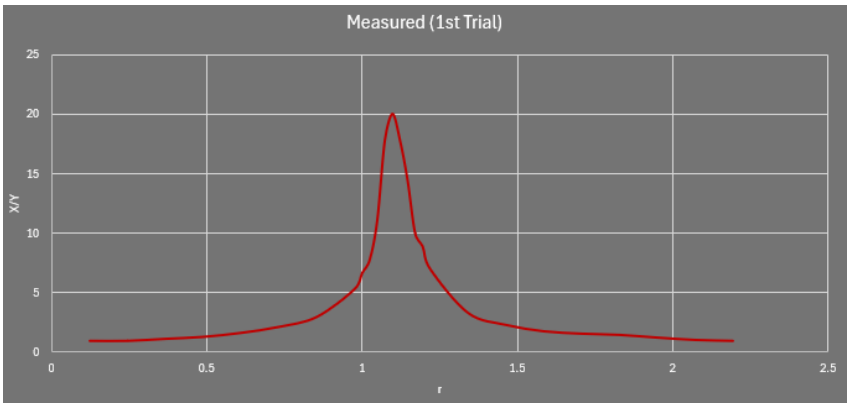
➢ Theoretical Values:

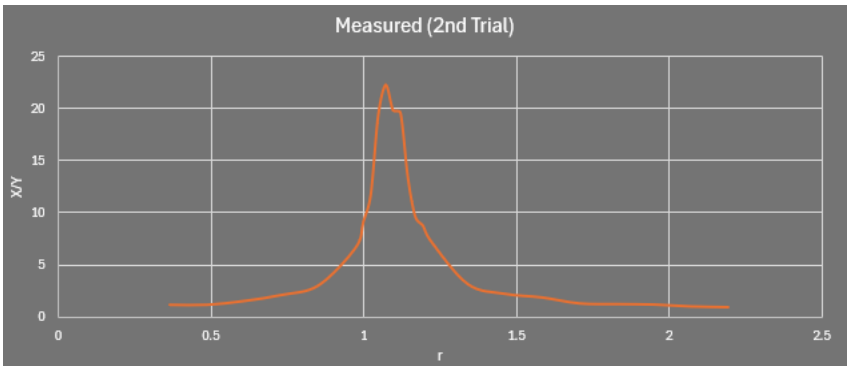
- $\omega_n = 61.906\ \text{rad/sec}$. Calculated using $\omega_n = \sqrt{\frac{k_{eq}}{m_{eq}}}$.
- ω is measured for each frequency (given in **Hz**) using $\omega = 2\pi f$

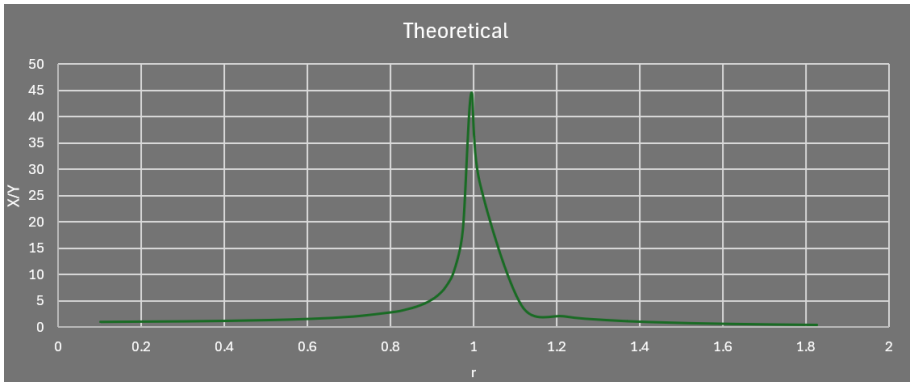
- $\boldsymbol{r}$ is calculated using the equation $\boldsymbol{r = \frac{\omega}{\omega_n}}$ for each $\boldsymbol{\omega}$.
- **Y** values were extracted by the Accelerometer for each frequency.
- **X** is calculated for each frequency using the given and calculated values for $\boldsymbol{\omega_n}$, $\boldsymbol{\omega}$, and **Y** by the equation $\boldsymbol{X = \frac{Y}{\sqrt{\left(1-\left(\frac{\omega}{\omega_n}\right)^2\right)^2+\left(2\zeta\frac{\omega}{\omega_n}\right)^2}}}$.
    - **(For the theoretical results and data, see appendix "*Forced Vibration theoretical Data*")**

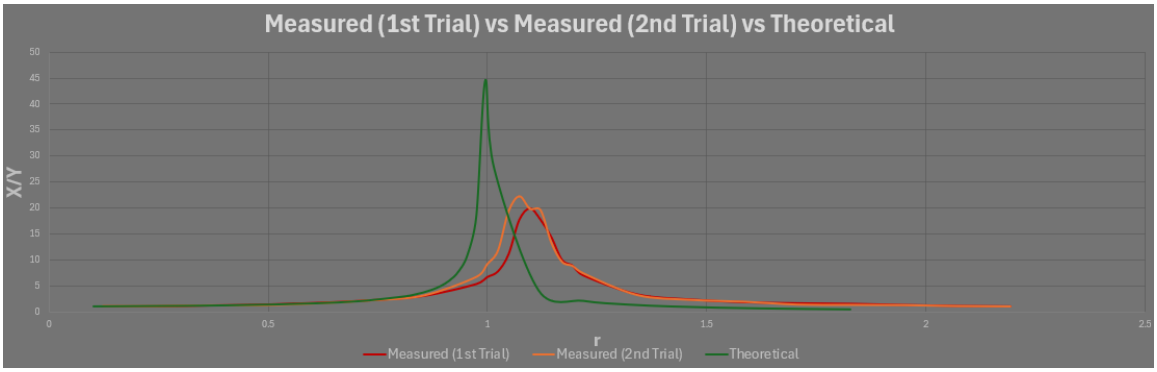
5. Plots

➢ The following equation, $\boldsymbol{\frac{X}{Y} = \frac{1}{\sqrt{(1-r^2)^2+(2\zeta r)^2}}}$, can be used to plot the measured and theoretical displacement amplitude ratios versus their corresponding frequency ratio.









6. Experimental and Theoretical Results Comparison

- The theoretical and experimental responses are aligned, as both are directly influenced by frequency and their corresponding natural frequency values.
- The experimental response **X** deviates slightly from theoretical predictions, particularly near resonance ($\mathbf{r \approx 1}$). This difference is due to imperfections, unaccounted damping, or accelerometer sensitivity.
- The theoretical data indicates a peak at $\mathbf{r \approx 1}$; however, the experimental peak is slightly shifted and displays a greater amplitude.
- Both datasets show a decline in **X** as $\mathbf{r > 1}$, but experimental values decay faster, likely due to higher energy dissipation in the real system.
- The measured natural frequency is about **16.6% lower** than the theoretical value.
- The experimental peak amplitude is **higher than predicted**, suggesting possible **nonlinear damping effects** or **energy amplification** due to system imperfections.
- Beyond resonance ($\mathbf{r > 1.1}$), experimental decays faster than theoretical values, suggesting additional damping mechanisms in the real system.

7. Comments on Results & Error Sources

- The natural frequency from the experimental results is slightly lower than the theoretical value because of the damping presence and energy dissipation as for the theoretical calculations idealized assumptions is considered such as uniform material properties and neglection of minor nonlinearities.
- The theoretical results provide more accurate approximation. Thus, it deviates from the experimental results because of the idealized assumptions.
- These differences focus on the effect of damping, friction, and material properties in real systems.

These discrepancies result from some source of errors in the real system which could be:

- **Measurement Errors:** errors of measuring the time intervals, displacement, and frequency using the experimental system.
- **Material and Structural Imperfections:** nonideal material properties like stiffness and damping. Any small difference in the setup impact in the setup vibration.
- **Energy Losses:** energy losses due to air resistance, friction, or internal material damping.
- **Excitation Variations:** inconsistency in amplitude and frequency of the forced function.
- The system was idealized to be a SDOF system.

Conclusion

In the end, we find that these experiments successfully achieved the fundamental principles of free vibration and forced vibration on the cantilever beam system. In the free vibration experiment, it provided us with the system's natural frequency, damping ratio, equivalent mass, and stiffness. In the second forced vibration experiment, it sheds light on the phenomenon of resonance and clarifies the relationship between displacement amplitude and excitation frequency. Comparison between experimental results and theoretical results shows minor deviation. Sources of error such as inaccurate measurements and environmental factors. These results highlight the importance of being very careful and controlling the conditions when studying vibration. This experiment illustrates important ideas about mechanical vibration and how it occurs in real life in an engineering system

References

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Appendices

Forced Vibration Measured Data Trial 1

Freq (Hz)	X = Beam Free end Acceleration (g)	Y = Beam Fixed end Acceleration (g)	frequency (ω)	natural frequency (ω_n)	r	$\frac{X}{Y}$
1	0.015	0.015	6.283185	51.6	0.121767	1
2	0.04	0.04	12.56637		0.243534	1
3	0.07	0.06	18.84956		0.365301	1.167
4	0.16	0.12	25.13274		0.487069	1.333
5	0.35	0.21	31.41593		0.608836	1.667
6	0.65	0.3	37.69911		0.730603	2.167
7	1.2	0.4	43.9823		0.85237	3
8	2.4	0.45	50.26548		0.974137	5.333
8.2	3	0.45	51.52212		0.998491	6.667
8.4	3.5	0.45	52.77876		1.022844	7.778
8.6	5	0.45	54.03539		1.047198	11.111
8.8	8	0.45	55.29203		1.071551	17.778
9	9	0.45	56.54867		1.095904	20
9.2	8	0.45	57.8053		1.120258	17.778
9.4	6.5	0.45	59.06194		1.144611	14.444
9.6	4.5	0.45	60.31858		1.168965	10
9.8	4	0.45	61.57522		1.193318	8.889
10	3.5	0.5	62.83185		1.217672	7
11	2.5	0.75	69.11504		1.339439	3.333
12	2.1	0.9	75.39822		1.461206	2.333
13	1.8	1	81.68141		1.582973	1.8
14	1.6	1	87.96459		1.70474	1.6
15	1.5	1	94.24778		1.826507	1.5
16	1.4	1.1	100.531		1.948275	1.273
17	1.4	1.3	106.8142		2.070042	1.077
18	1.4	1.4	113.0973		2.191809	1

Forced Vibration Measured Data Trial 2

Freq (Hz)	X = Beam Free end Acceleration (g)	Y = Beam Fixed end Acceleration (g)	frequency (ω)	natural frequency (ω_n)	r	$\frac{X}{Y}$
1	0.015	0.015	6.283185307	51.6	0.121767	1
2	0.035	0.035	12.56637061		0.243534	1
3	0.075	0.065	18.84955592		0.365301	1.154
4	0.15	0.13	25.13274123		0.487069	1.154
5	0.35	0.23	31.41592654		0.608836	1.522
6	0.65	0.31	37.69911184		0.730603	2.097
7	1.3	0.43	43.98229715		0.85237	3.023
8	3	0.45	50.26548246		0.974137	6.667
8.2	4.1	0.45	51.52211952		0.998491	9.111
8.4	5.5	0.47	52.77875658		1.022844	11.702
8.6	8.7	0.45	54.03539364		1.047198	19.333
8.8	10	0.45	55.2920307		1.071551	22.222
9	9.1	0.46	56.54866776		1.095904	19.783
9.2	8.2	0.42	57.80530483		1.120258	19.524
9.4	5.7	0.43	59.06194189		1.144611	13.256
9.6	4.3	0.45	60.31857895		1.168965	9.556
9.8	3.5	0.4	61.57521601		1.193318	8.75
10	3.3	0.45	62.83185307		1.217672	7.333
11	2.05	0.65	69.11503838		1.339439	3.154
12	1.9	0.86	75.39822369		1.461206	2.209
13	1.75	0.94	81.68140899		1.582973	1.862
14	1.55	1.2	87.9645943		1.70474	1.292
15	1.35	1.1	94.24777961		1.826507	1.227
16	1.3	1.1	100.5309649		1.948275	1.182
17	1.2	1.2	106.8141502		2.070042	1
18	1.3	1.4	113.0973355		2.191809	0.929

Forced Vibration Theoretical Data

Freq (Hz)	frequency (ω_n)	natural frequency (ω_n)	r	Y = Beam Fixed end Acceleration (g)	Damping Ratio (ζ)	X	$\frac{X}{Y}$
1	6.28319	61.906	0.1015	0.015	0.009964	0.015	1
2	12.56637		0.20299	0.035		0.037	1.057
3	18.84956		0.30449	0.065		0.072	1.108
4	25.13274		0.40598	0.13		0.156	1.2
5	31.41593		0.50748	0.23		0.31	1.348
6	37.69911		0.60897	0.31		0.493	1.59
7	43.9823		0.71047	0.43		0.868	2.019
8	50.26548		0.81196	0.45		1.319	2.931
8.2	51.52212		0.83226	0.45		1.462	3.249
8.4	52.77876		0.85256	0.47		1.717	3.653
8.6	54.03539		0.87286	0.45		1.885	4.189
8.8	55.29203		0.89316	0.45		2.216	4.924
9	56.54867		0.91346	0.46		2.761	6.002
9.2	57.8053		0.93376	0.42		3.245	7.726
9.4	59.06194		0.95406	0.43		4.686	10.898
9.6	60.31858		0.97436	0.45		8.3	18.444
9.8	61.57522		0.99466	0.4		17.776	44.44
10	62.83185		1.01496	0.45		12.396	27.547
11	69.11504		1.11645	0.65		2.627	4.042
12	75.39822		1.21795	0.86		1.777	2.066
13	81.68141		1.31944	0.94		1.268	1.349
14	87.96459		1.42094	1.2		1.177	0.981
15	94.24778		1.52243	1.1		0.835	0.759
16	100.53096		1.62393	1.1		0.672	0.611
17	106.81415		1.72542	1.2		0.607	0.506
18	113.09734		1.82692	1.4		0.599	0.428